

Deliverable 1

What is a web server? (In the context of software Ex. Apache)

- *A web server is software that receives requests from a web browser and responds by sending back web pages, images, or other files. It is used to host websites and deliver content to users over a network.*

What are some different web server applications? Include definitions, project's website/where to download it, which operating system is available for and its latest version.

- Web server applications include:
 - **Apache HTTP Server** (<https://httpd.apache.org>)
 - *Apache is a free, open-source, cross-platform web server used to serve websites over HTTP/HTTPS.*
 - *Latest Version: Apache HTTP Server 2.4.67*
 - *Available for Linux, Windows, and macOS*
 - **Nginx** (<https://nginx.org>)
 - *Nginx is a high-performance web server known for speed, low resource usage, and its ability to act as a reverse proxy and load balancer.*
 - *Latest Version: Nginx 1.30.0*
 - *Available for Linux and Windows*
 - **Lighttpd** (<https://www.lighttpd.net>)
 - *Lighttpd is a lightweight, fast web server optimized for speed and low memory usage.*
 - *Latest Version: Lighttpd 1.4.82*
 - *Available for Linux and macOS*

What is virtualization?

- *Virtualization is defined as creating virtual versions of something.*

What is virtualbox?

- *Virtualbox is an open-source virtualization program that allows you to create virtual machines.*

What is a virtual machine?

- *A virtual machine is a simulated computer created by a virtualization program.*

In the context of virtualization, what does host machine and guest machine mean?

- *Host machine refers to the computer where the hypervisor is installed. This is the virtualization program you used to create your machine. Guest machine refers to the computer being simulated in the virtual machine.*

What is Debian?

- *Debian is one of the 2 main Linux distributions. Most distributions are branches of either Debian or Redhat.*

What is a firewall?

- *A firewall is software that filters network traffic. It decides which connections are allowed and which are blocked to protect the system.*

What is SSH?

- *SSH (Secure Shell) is a secure protocol used to remotely access and manage a server.*

What is an IP Address?

- *An IP Address is a numerical label assigned to a device on a network so it can communicate with other devices.*

What is a network mask?

- *A network mask is a number that tells a computer which part of an IP address refers to the network and which part refers to the device.*

What is a port? (in the context of networking/computers)

- *A port is a virtual communication endpoint used by programs to send and receive data.*

What is port forwarding?

- *Port forwarding allows you to map a port on the host machine to a port on the guest machine. This allows outside devices to access services running inside the guest machine.*

What is localhost? (in the context of networking/computers)

- *Localhost is the hostname that refers to your own computer.*

What does this ip address represent 127.0.0.1?

- *This ip address represents your local machine.*

What is Git?

- *Git is a version control system that tracks changes in files so developers can manage your projects.*

What is GitHub?

- *GitHub is an online platform that hosts Git repositories. It allows you to store code, share projects, and collaborate with others.*